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ROLL NUMBER:

EXERCISE 6

DELUGE

1. Aim

To study and implement basic Deluge scripting concepts such as variables, data type conversion, arithmetic expressions, relational and logical operators, and conditional statements in Zoho Creator.

2. Introduction

Deluge is the scripting language used in Zoho applications to automate tasks and perform logical operations. It allows users to work with variables, perform calculations, compare values, apply logical conditions, and control the flow of a program. In this exercise, simple Deluge scripts are used to perform calculations, validate values, check conditions, and store results.

3. Objectives

- To understand the basic syntax of Deluge scripting.
- To declare and use different types of variables.
- To perform arithmetic and relational operations.
- To use logical operators such as AND (&&) and OR (||).
- To implement if-else conditions in Deluge.
- To convert values from one data type to another.
- To display and verify the results of Deluge expressions.

4. DELUGE Codes & Outputs

The screenshot shows a user interface for a Deluge script exercise. At the top, it says "Data Types" and "MEDIUM". The task is: "Declare a Date variable `dob` using the `toDate` function on the text `"01-Jan-2004"` .". Below this, there is a section "Your Deluge Script" with a code editor containing the following script:

```
dob = "01-Jan-2004".toDate()
info dob;
```

Below the code editor, there is a green success message: "Correct! +1.6 points. Great job on Data Types." and a "Model answer" showing the full script: `dob = "01-Jan-2004".toDate("dd-MMM-yyyy");`. At the bottom, there is a "Next Question →" button.

Conditional Statements

MEDIUM

Write an if-else statement that checks whether `stock` is equal to 0, setting `availability` to "Out of Stock", else "Available" .

Your Deluge Script

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```
{
  Available="Out of Stock";
}
else
{
  Available="availability";
}
```

✓ **Correct!** +1.6 points. Great job on **Conditional Statements**.

Model answer: `if(stock == 0) { availability = "Out of Stock"; } else { availability = "Available"; }`

Expressions

MEDIUM

Write an expression using parentheses to correctly compute `(a + b) * c` , storing the result in `output` .

Your Deluge Script

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```
output=(a+b)*c
```

✓ **Correct!** +1.6 points. Great job on **Expressions**.

Model answer: `output = (a + b) * c;`

Comments

EASY

Write a single-line comment in Deluge explaining: `this calculates total marks` .

Your Deluge Script

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```
total marks=100; //this calculates total marks
```

✓ **Correct!** +1.1 points. Great job on **Comments**.

Model answer: `// this calculates total marks`

Data Types

HARD

Convert the number variable `marks` (holding 90) into Text data type and store it in `marksStr` .

Your Deluge Script

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```
marks=90;
marksStr=marks.toString();
```

✅ **Correct!** +2.7 points. Great job on **Data Types**.

Model answer: `marksStr = marks.toString();`

Variables

HARD

Declare variables `principal` = 1000 and `rate` = 5, then use them in an expression to compute `interest` as $(\text{principal} * \text{rate}) / 100$.

Your Deluge Script

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```
principal=1000;
rate=5;
interest=(principal*rate)/100;
```

✅ **Correct!** +2.7 points. Great job on **Variables**.

Model answer: `principal = 1000; rate = 5; interest = (principal * rate) / 100;`

Expressions

MEDIUM

Write an expression that combines a relational and logical operator to check if `marks` is between 40 and 100 (inclusive), storing result in `isValid` .

Your Deluge Script

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```
isValid=(marks>=40) && (marks<=100);
```

✅ **Correct!** +1.6 points. Great job on **Expressions**.

Model answer: `isValid = (marks >= 40 && marks <= 100);`

Data Types

MEDIUM

Declare a Boolean variable `hasScholarship` set to `false`.

Your Deluge Script

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```
hasScholarship=false;
```

✅ **Correct!** +1.6 points. Great job on **Data Types**.

Model answer: `hasScholarship = false;`

Conditional Statements

MEDIUM

Write an if statement using AND (`&&`) that sets `eligible` to `true` only when both `attendance` is above 75 and `marks` is above 50.

Your Deluge Script

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```
if(attendance>75 && marks>50)
{
    eligible=true;
}
```

✅ **Correct!** +1.6 points. Great job on **Conditional Statements**.

Model answer: `if(attendance > 75 && marks > 50) { eligible = true; }`

Operators

MEDIUM

Write a logical expression using OR (`||`) that checks whether `day` equals `"Saturday"` or `"Sunday"`, storing result in `isWeekend`.

Your Deluge Script

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```
isWeekend =(day=="Saturday" || day=="Sunday");
```

✅ **Correct!** +1.6 points. Great job on **Operators**.

Model answer: `isWeekend = (day == "Saturday" || day == "Sunday");`

5. Advantages

- Simple syntax makes basic scripts easy to write and understand.
- Supports variables, expressions, conditions, and logical operations.
- Helps automate repetitive tasks in Zoho applications.
- Provides built-in functions for common operations and data conversion.
- Can be used to validate and process application data efficiently.

6. Disadvantages

- Deluge is mainly designed for the Zoho ecosystem and has limited use outside it.
- Complex scripts can become difficult to maintain without proper organization.
- Advanced requirements may need knowledge of Zoho Creator and its application structure.
- Debugging complicated scripts can require careful testing.
- The available scripting features depend on the Zoho application being used.

7. Result

The Deluge scripting exercise was completed successfully. Basic variables, arithmetic expressions, data type conversion, relational and logical operators, and conditional statements were implemented and tested. The expected results were obtained for the given scripts.